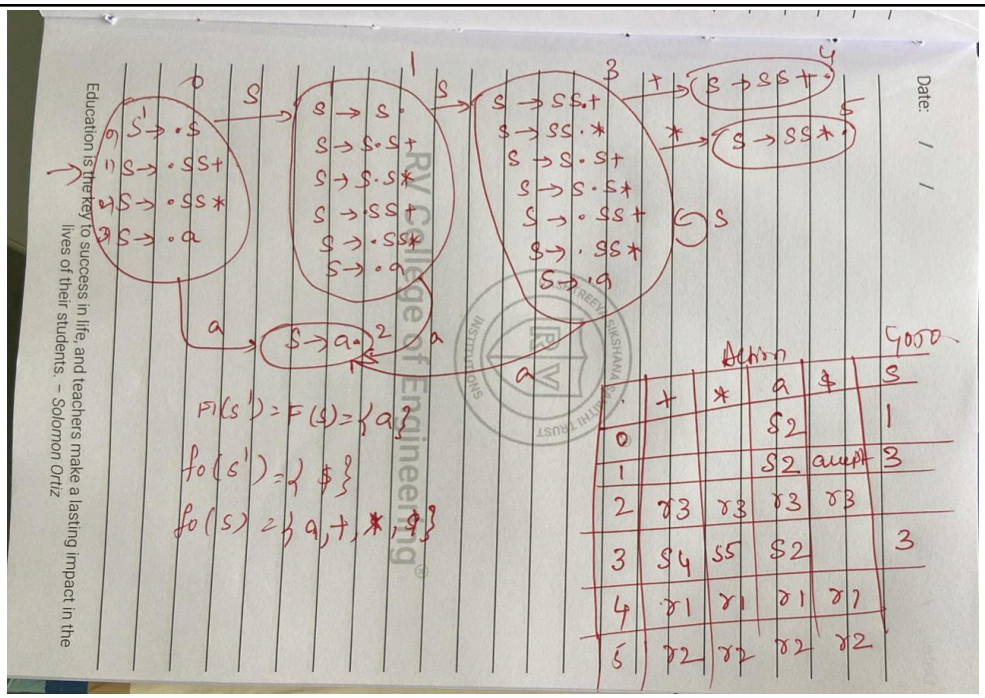
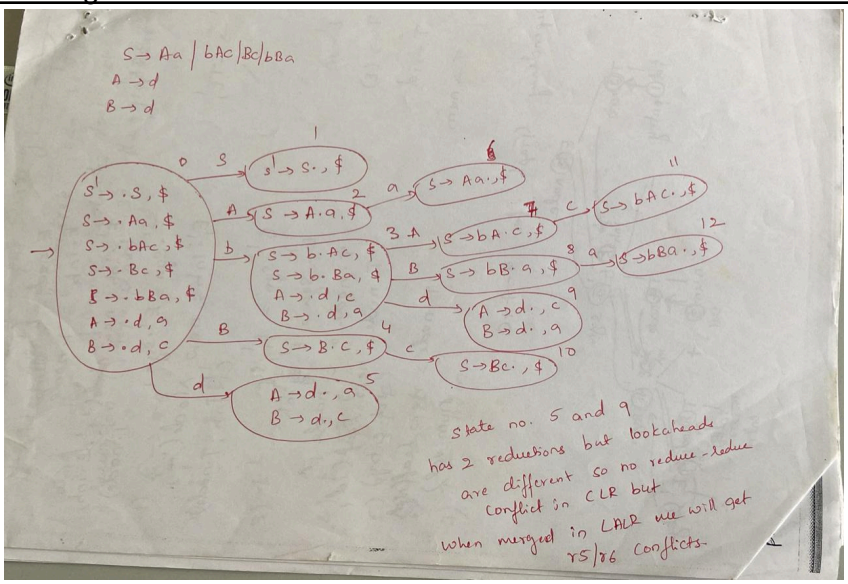
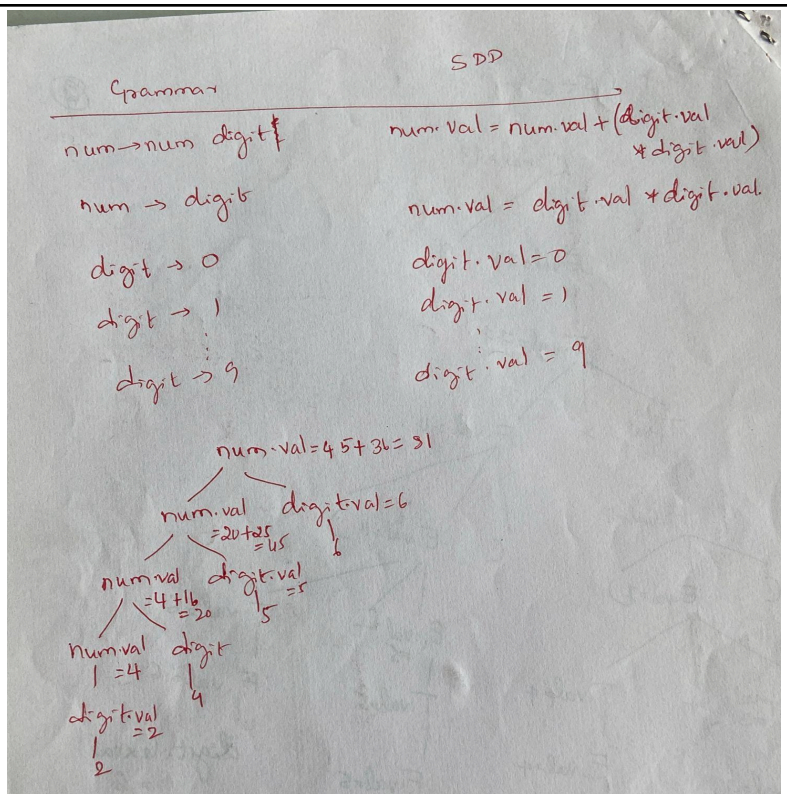
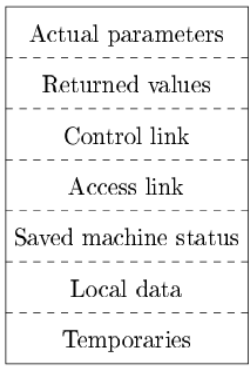
	R V College of Engineering **Department of Computer Science and Engineering** **CIE - II: Question Paper**		
Subject : (Code)	**COMPILER DESIGN (CS363AI)**	**Semester : 6th BE**	
Date : JUNE 2025	**Duration : 120** minutes	**Staff : DD/SB/SNM/SMS**	
Name :	**USN** :	**Section : A,B,C,D**	

Si. No	**Part A** **Quiz Questions**	**M**	**BT**	**Co**
1.	What is the meaning of LR(k) grammar. Give an example. Reading from left, applying RMD in reserve order looking k symbols ahead. --- 1M Example -1M	2		
2.	Consider the grammar below: S->SS+ | SS* | a If I ={S->S.S+} compute Closure of (I). Closure(I)={ S->S.S+, S->.SS+, S->.SS*, S->.a}	2		
3.	Consider the grammar below: S->bAc | dC| bdA A->d C->k Compute the Canonical set of LR(1) collections for the starting state. S'->.S,$ S->.bAc,$ | .dC, $| .bdA, $	2		
4.	What is an L-attributed SDD? Give an example. L- Attributed SDD can have both synthesized and inherited attributes for the nodes of the annotated parse tree. –1 M Example—1M	2		
5.	What is the relationship between *activation record* and *control stack*? Procedure calls and returns are usually managed by run-time stack called the control stack . Each live activation has an activation record on the control stack.	1		
6.	List the properties we desire of memory manager. Space efficiency, program efficiency, low overhead	1		
	Part B			

1.	 LR(0) automata – 4 Marks Parsing table – 3 Marks Parsing – 3Marks			
2	 LR(1) automata – 4marks Parsing table construction for CLR(1) or LALR(1) – 4marks Justification for not LALR – 2 marks			
3	SDD --- 5 marks Annotated parse tree – 5marks			

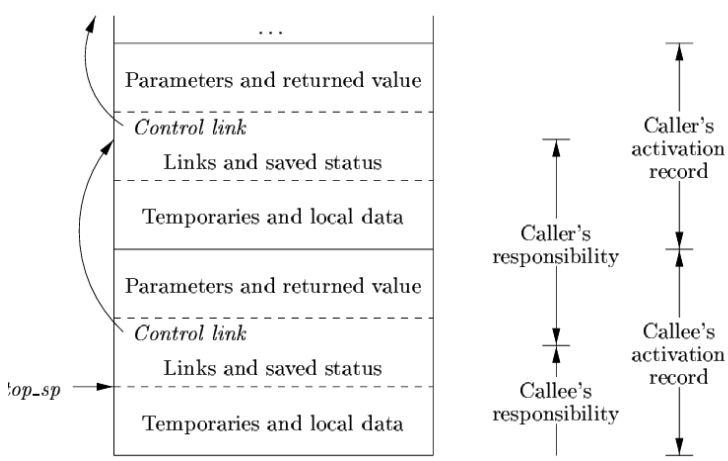


4a.	Any one of the following: Creating syntax tree or defining structure of a type with sematic rules and an example 3+2 = 5marks			
4b.				

1. Temporary values, such as those arising from the evaluation of expressions, in cases where those temporaries cannot be held in registers.
2. Local data belonging to the procedure whose activation record this is.
3. A saved machine status, with information about the state of the machine just before the call to the procedure. This information typically includes the *return address* (value of the program counter, to which the called procedure must return) and the contents of registers that were used by the calling procedure and that must be restored when the return occurs.
4. An "access link" may be needed to locate data needed by the called procedure but found elsewhere, e.g., in another activation record. Access links are discussed in Section 7.3.5.
5. A *control link*, pointing to the activation record of the caller.
6. Space for the return value of the called function, if any. Again, not all called procedures return a value, and if one does, we may prefer to place that value in a register for efficiency.
7. The actual parameters used by the calling procedure. Commonly, these values are not placed in the activation record but rather in registers, when possible, for greater efficiency. However, we show a space for them to be completely general.

Diagram + explanation = 2+3 = 5

5. Design principles:
   1. Values communicated between caller and callee are generally placed at the beginning of the callee's activation record, so they are as close as possible to the caller's activation record.
   2. Fixed-length items are generally placed in the middle.
   3. Items whose size may not be known early enough are placed at the end of the activation record.
   4. Wemust locate the top-of-stack pointer judiciously. A common approach is to haveitpointtotheend of the xed-length elds in the activation record.



Design principles – 4 marks
Diagram – 2 marks
Explanation – 4Marks